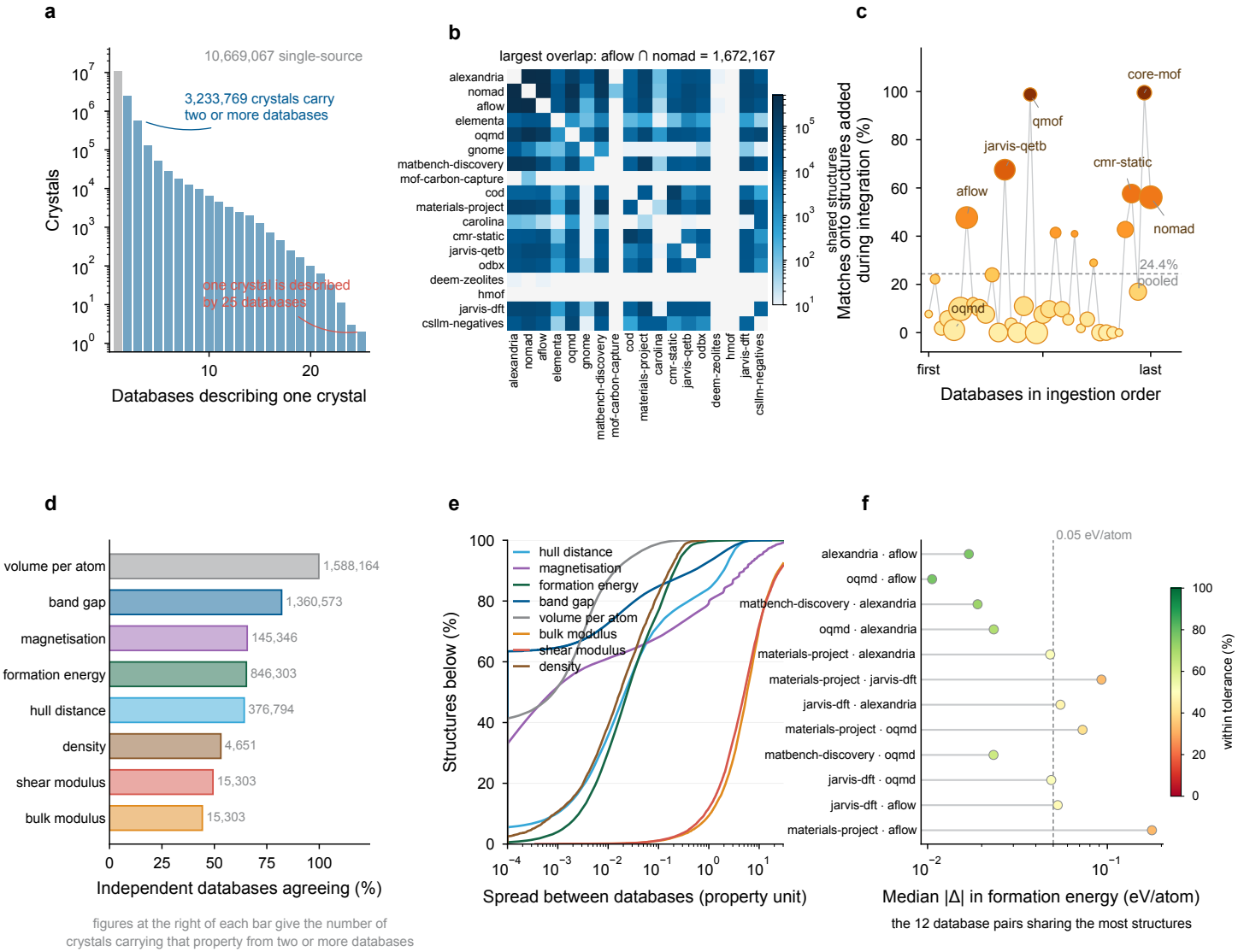

a
10,669,067 single-source
3,233,769 crystals carry two or more databases
one crystal is described by 25 databases
Crystals
Databases describing one crystal
b
largest overlap: aflow ∩ nomad = 1,672,167
alexandria
nomad
aflow
elementa
oqmd
gnome
matbench-discovery
mof-carbon-capture
cod
materials-project
carolina
cmr-static
jarvis-qetb
odbx
deem-zeolites
hmof
jarvis-dft
csllm-negatives
shared structures
c
Matches onto structures added during integration (%)
core-mof
qmof
jarvis-qetb
cmr-static
aflow
nomad
24.4% pooled
oqmd
first
last
Databases in ingestion order
d
volume per atom
band gap
magnetisation
formation energy
hull distance
density
shear modulus
bulk modulus
1,588,164
1,360,573
145,346
846,303
376,794
4,651
15,303
15,303
Independent databases agreeing (%)
figures at the right of each bar give the number of crystals carrying that property from two or more databases
e
hull distance
magnetisation
formation energy
band gap
volume per atom
bulk modulus
shear modulus
density
Structures below (%)
Spread between databases (property unit)
f
0.05 eV/atom
alexandria · aflow
oqmd · aflow
matbench-discovery · alexandria
oqmd · alexandria
materials-project · alexandria
materials-project · jarvis-dft
jarvis-dft · alexandria
materials-project · oqmd
matbench-discovery · oqmd
jarvis-dft · oqmd
jarvis-dft · aflow
materials-project · aflow
within tolerance (%)
Median |Δ| in formation energy (eV/atom)
the 12 database pairs sharing the most structures